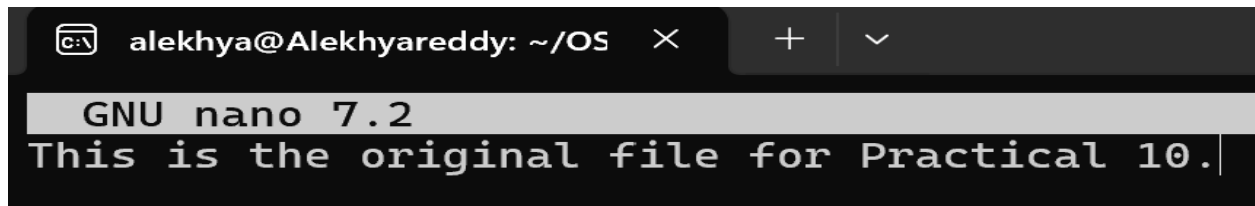


PRACTICAL WEEK-10

1 . Investigate inode structures using commands such as `ls -i`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

Aim : To study inode structures in Linux and understand the difference between hard links and symbolic links.

Program :

A screenshot of a terminal window. The top bar shows the user 'alekhya@AlekhyaReddy' in the directory '~/OS'. Below the bar, the text 'GNU nano 7.2' is displayed. The main content of the terminal shows the text 'This is the original file for Practical 10.' with a cursor at the end of the line.

```
alekhya@AlekhyaReddy: ~/OS × + ∨  
GNU nano 7.2  
This is the original file for Practical 10.
```

Output :

alekhya@Alekhyareddy: ~/OS X + v

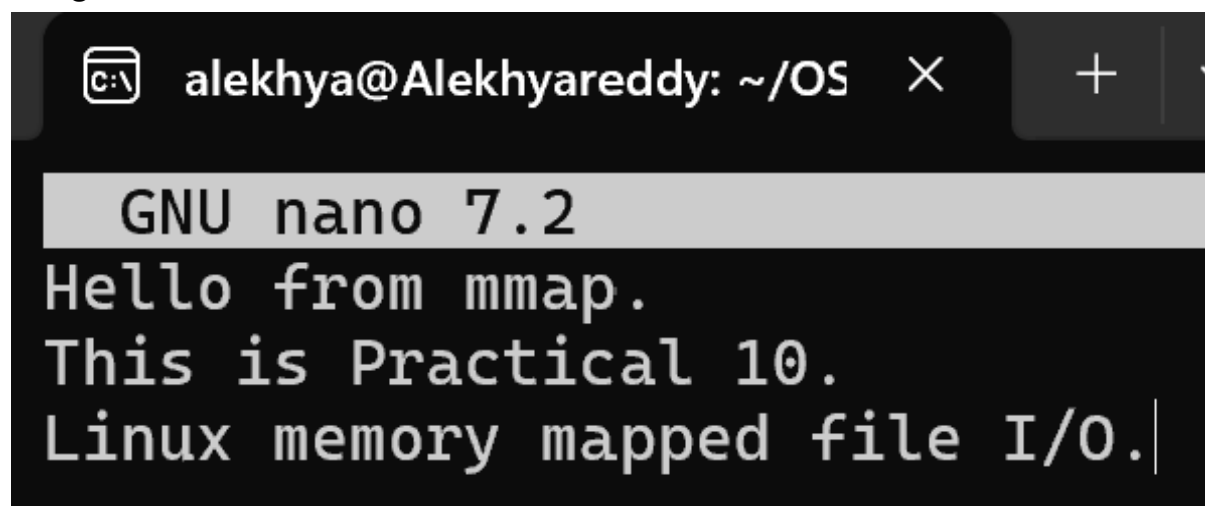
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost proxies.

```
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano original.txt
alekhya@Alekhyareddy:~/OSSP$ ls -li original.txt
63450 original.txt
alekhya@Alekhyareddy:~/OSSP$ ls -li original.txt
63450 -rw-r--r-- 1 alekhya alekhya 44 Sep 23 05:34 original.txt
alekhya@Alekhyareddy:~/OSSP$ stat original.txt
  File: original.txt
  Size: 44          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 63450        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ alekhya)   Gid: ( 1000/ alekhya)
Access: 2026-09-23 05:34:08.453148705 +0000
Modify: 2026-09-23 05:34:08.456790042 +0000
Change: 2026-09-23 05:34:08.456790042 +0000
 Birth: 2026-09-23 05:34:08.453148705 +0000
alekhya@Alekhyareddy:~/OSSP$ ls -li original.txt
63450 original.txt
alekhya@Alekhyareddy:~/OSSP$ find . -inum 1785
alekhya@Alekhyareddy:~/OSSP$ ln original.txt hardlink.txt
alekhya@Alekhyareddy:~/OSSP$ ls -li original.txt hardlink.txt
63450 -rw-r--r-- 2 alekhya alekhya 44 Sep 23 05:34 hardlink.txt
63450 -rw-r--r-- 2 alekhya alekhya 44 Sep 23 05:34 original.txt
alekhya@Alekhyareddy:~/OSSP$ stat original.txt
  File: original.txt
  Size: 44          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 63450        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ alekhya)   Gid: ( 1000/ alekhya)
Access: 2026-09-23 05:34:08.453148705 +0000
Modify: 2026-09-23 05:34:08.456790042 +0000
Change: 2026-09-23 05:36:40.136141249 +0000
 Birth: 2026-09-23 05:34:08.453148705 +0000
alekhya@Alekhyareddy:~/OSSP$ ln -s original.txt symlink.txt
alekhya@Alekhyareddy:~/OSSP$ ls -li original.txt hardlink.txt symlink.txt
63450 -rw-r--r-- 2 alekhya alekhya 44 Sep 23 05:34 hardlink.txt
63450 -rw-r--r-- 2 alekhya alekhya 44 Sep 23 05:34 original.txt
39887 lrwxrwxrwx 1 alekhya alekhya 12 Sep 23 05:37 symlink.txt -> original.txt
alekhya@Alekhyareddy:~/OSSP$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48   Inode: 39887        Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ alekhya)   Gid: ( 1000/ alekhya)
Access: 2026-09-23 05:37:08.209324364 +0000
Modify: 2026-09-23 05:37:02.868784478 +0000
Change: 2026-09-23 05:37:02.868784478 +0000
 Birth: 2026-09-23 05:37:02.868784478 +0000
alekhya@Alekhyareddy:~/OSSP$ ls -l symlink.txt
lrwxrwxrwx 1 alekhya alekhya 12 Sep 23 05:37 symlink.txt -> original.txt
alekhya@Alekhyareddy:~/OSSP$ ls -li symlink.txt
39887 symlink.txt
alekhya@Alekhyareddy:~/OSSP$ find . -inum INODE_NUMBER
find: invalid argument 'INODE_NUMBER' to '-inum'
alekhya@Alekhyareddy:~/OSSP$ rm -f original.txt hardlink.txt symlink.txt
alekhya@Alekhyareddy:~/OSSP$ |
```

2 . Design a program using `mmap()` to perform file reading and writing. Compare memory-mapped I/O with traditional `read()/write()` operations in terms of efficiency and implementation complexity.

Aim : To implement file reading and writing using `mmap()` and compare it with traditional `read()/write()` operations.

Program :

A screenshot of a terminal window. The title bar shows the user 'alekhya@AlekhyaReddy' in the directory '~/OS'. The terminal content shows the 'GNU nano 7.2' prompt, followed by three lines of text: 'Hello from mmap.', 'This is Practical 10.', and 'Linux memory mapped file I/O.' with a cursor at the end of the last line.

```
alekhya@AlekhyaReddy: ~/OS  
GNU nano 7.2  
Hello from mmap.  
This is Practical 10.  
Linux memory mapped file I/O.
```



alekhyareddy: ~/OS



GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    int fd;
    struct stat st;
    char *data;

    fd = open("mmap_file.txt", O_RDWR);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size,
                PROT_READ | PROT_WRITE,
                MAP_SHARED, fd, 0);

    if (data == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original content:\n");
    write(STDOUT_FILENO, data, st.st_size);

    memcpy(data, "Modified using mmap().\n", 23);

    if (msync(data, st.st_size, MS_SYNC) == -1)
    {
        perror("msync");
    }

    printf("\nFile modified successfully using mmap().\n");

    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```



alekhya@Alekhyareddy: ~/OS



GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main()
{
    int fd;
    ssize_t bytes_read;
    char buffer[BUFFER_SIZE];

    fd = open("mmap_file.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    bytes_read = read(fd, buffer, BUFFER_SIZE - 1);

    if (bytes_read == -1)
    {
        perror("read");
        close(fd);
        return 1;
    }

    buffer[bytes_read] = '\0';

    printf("Content using read():\n");
    printf("%s", buffer);

    close(fd);

    return 0;
}
```

Output :

```
alekhya@Alekhyaireddy: ~/OS  +  v
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost proxies.
alekhya@Alekhyaireddy:~$ cd ~/OSSP
alekhya@Alekhyaireddy:~/OSSP$ nano mmap_file.txt

alekhya@Alekhyaireddy:~/OSSP$ cat mmap_file.txt
Hello from mmap.
This is Practical 10.
Linux memory mapped file I/O.
alekhya@Alekhyaireddy:~/OSSP$ nano mmap_file.c
alekhya@Alekhyaireddy:~/OSSP$ gcc -Wall -Wextra mmap_file.c -o mmap_file
alekhya@Alekhyaireddy:~/OSSP$ ./mmap_file
Original content:
Hello from mmap.
This is Practical 10.
Linux memory mapped file I/O.

File modified successfully using mmap().
alekhya@Alekhyaireddy:~/OSSP$ cat mmap_file.txt
Modified using mmap().
s Practical 10.
Linux memory mapped file I/O.
alekhya@Alekhyaireddy:~/OSSP$ nano traditional_io.c
alekhya@Alekhyaireddy:~/OSSP$ gcc -Wall -Wextra traditional_io.c -o traditional_io
alekhya@Alekhyaireddy:~/OSSP$ ./traditional_io
Content using read():
Modified using mmap().
s Practical 10.
Linux memory mapped file I/O.
alekhya@Alekhyaireddy:~/OSSP$ time ./mmap_file
Original content:
Modified using mmap().
s Practical 10.
Linux memory mapped file I/O.

File modified successfully using mmap().

real    0m0.041s
user    0m0.002s
sys     0m0.000s
alekhya@Alekhyaireddy:~/OSSP$ time ./traditional_io
Content using read():
Modified using mmap().
s Practical 10.
Linux memory mapped file I/O.

real    0m0.002s
user    0m0.000s
sys     0m0.000s
alekhya@Alekhyaireddy:~/OSSP$
```